Reduced-order modelling of a cello body transfer function: First Report

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June 25, 2026

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1 Project specification

The overall aim of this project is to effectively model and emulate the characteristics of an acoustic string instrument body via excitation of the instrument bridge with an impulse-like signal. This can be applied in the following scenarios:

- Conversion of the bridge force to sound pressure: a transfer function can be retrieved to translate the bridge force signal to an acoustic pressure signal. Applications include electric and 3D printed cellos and virtual-acoustic bowed-string instruments and pedal effects.
- Bridge force to bridge velocity: deriving the bridge admittance that needs to be known in modelling and simulation of bowed-string vibrations.

Cello bridge admittance features fewer modes than the equivalent transfer function, thus is much simpler to model.

An impulse response of the system can be decomposed into its modal form, which yields a parallel set of 2nd-order IIR filters. This is equivalent to modelling each mode as a decaying harmonic oscillator. The aim of the project is to employ the Eigensystem Realisation Algorithm (ERA) to achieve this decomposition.

This method is potentially useful due to the singular value decomposition stage employed in the algorithm, which will allow a reduction in the order of the modes by removing singular values (and associated left and right singular vectors) from the system.

The expected tasks of the project as as follows:

1. Develop an impact hammer system with which repeatable measurements can be conducted;

*Repository: github.com/Gijinkstra/estimation-theory-car-tracker/tree/recover-branch

2. Apply this to experimentally acquire the IRs of a number of cello bodies';
3. Use the ERA to modally decompose mono and stereo IRs into a reduced-order sum of body modes;
4. Test and evaluate the ERA, comparing performance to the Filter Diagonalisation Method (FDM).
5. Create a set of cello body models (in modally decomposed form) and develop demos that exemplify these.

The expected learning outcomes of the project are:

1. A thorough understanding of modal decomposition methods.
2. Developed skills for acoustic measurement and audio signal acquisition.
3. Practical knowledge on acoustic impulse response applications.
4. Developed necessary programming skills.